

Probability
Chapter 2 - Module 3

Joint Probability Distributions

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One way to determine the joint probability distribution is to use the definition of the cumulative distribution function and find its derivatives.

This would obviously be a difficult process in most cases and we need additional tools.

Let X_1 and X_2 be random variables with joint pdf $f(x_1, x_2)$.

Let

$$y_1 = g_1(x_1, x_2) \text{ and } y_2 = g_2(x_1, x_2)$$

and define

$$Y_1 = g_1(X_1, X_2) \text{ and } Y_2 = g_2(X_1, X_2).$$

What is the joint distribution of Y_1 and Y_2 ?

We make two assumptions.

1. The equations $y_1 = g_1(x_1, x_2)$ and $y_2 = g_2(x_1, x_2)$ can be solved for x_1 and x_2 to give new functions $x_1 = h_1(y_1, y_2)$ and $x_2 = h_2(y_1, y_2)$.
2. The Jacobian satisfies

$$J(x_1, x_2) \equiv \begin{vmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \end{vmatrix} = \frac{\partial g_1}{\partial x_1} \frac{\partial g_2}{\partial x_2} - \frac{\partial g_1}{\partial x_2} \frac{\partial g_2}{\partial x_1} \neq 0$$

for all (x_1, x_2) .

Theorem. If assumptions 1 and 2 are satisfied, then

$$f_{Y_1, Y_2}(y_1, y_2) = \begin{cases} f_{X_1, X_2}(x_1, x_2) |J(x_1, x_2)|^{-1} & \text{if } y_1 = g_1(x_1, x_2) \text{ and } y_2 = g_2(x_1, x_2) \\ & \text{for some } x_1 \text{ and } x_2 \\ 0 & \text{otherwise,} \end{cases}$$

where $x_1 = h_1(y_1, y_2)$ and $x_2 = h_2(y_1, y_2)$.

Example 1. We return to the previous case where we considered the sum and difference of two uniform random variables.

So let X_1 and X_2 be two independent uniform $[0, 1]$ random variables. And let

$$y_1 = g_1(x_1, x_2) \equiv x_1 + x_2 \text{ and } y_2 = g_2(x_1, x_2) = x_1 - x_2$$

so that

$$Y_1 = g_1(X_1, X_2) = X_1 + X_2 \text{ and } Y_2 = g_2(X_1, X_2) = X_1 - X_2.$$

It is easy to see that the Jacobian is

$$J(x_1, x_2) = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$$

and so $|J(x_1, x_2)| = 2$.

So

$$\begin{aligned} f_{Y_1, Y_2}(y_1, y_2) &= f_{X_1, X_2}(x_1, x_2) |J(x_1, x_2)|^{-1} \\ &= \frac{1}{2} f_{X_1, X_2} \left(\frac{y_1 + y_2}{2}, \frac{y_1 - y_2}{2} \right). \end{aligned}$$

$$\begin{aligned}
 f_{Y_1, Y_2}(y_1, y_2) &= \frac{1}{2} f_{X_1, X_2} \left(\frac{y_1 + y_2}{2}, \frac{y_1 - y_2}{2} \right) \\
 &= \begin{cases} \frac{1}{2} & 0 \leq y_1 + y_2 \leq 2 \text{ and } 0 \leq y_1 - y_2 \leq 2 \\ 0 & \text{otherwise.} \end{cases}
 \end{aligned}$$

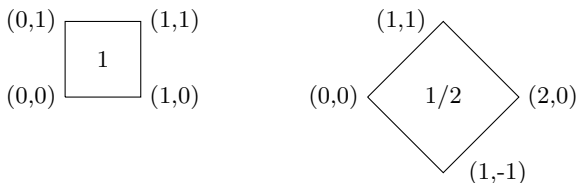
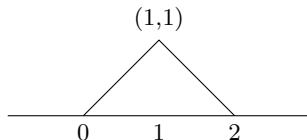


Figure 1. Original unit square and rotated and scaled square.

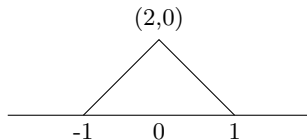
This appears to be correct since

$$f_{Y_1}(y_1) = \int_{-\infty}^{\infty} f_{Y_1, Y_2}(y_1, y_2) dy_2 =$$



and

$$f_{Y_2}(y_2) = \int_{-\infty}^{\infty} f_{Y_1, Y_2}(y_1, y_2) dy_1 =$$



Question: So are Y_1 and Y_2 independent.

Answer: No, because $f_{Y_1, Y_2}(y_1, y_2) = 1/2$ and is constant and $f_{Y_1}(y_1)f_{Y_2}(y_2)$ is not!

These ideas naturally extend to more than two variables in the following way.

Theorem. Let $X_1, X_2, \dots, X_n$ be random variables with pdf $f(x_1, x_2, \dots, x_n)$. And let

$$y_1 = g_1(x_1, x_2, \dots, x_n), y_2 = g_2(x_1, x_2, \dots, x_n), \dots, y_n = g_n(x_1, x_2, \dots, x_n),$$

or more compactly $y = g(x)$, where

$$x = (x_1, x_2, \dots, x_n)^T,$$

$$y = (y_1, y_2, \dots, y_n)^T$$

and

$$g(x) = (g_1(x), g_2(x), \dots, g_n(x))^T$$

Assumption 1. The n equations in n unknowns $y = g(x)$ can be solved for x to give the inverse function $x = h(y)$.

Assumption 2. The Jacobian

$$J(x) = J(x_1, x_2, \dots, x_n) = \begin{vmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \dots & \frac{\partial g_1}{\partial x_n} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \dots & \frac{\partial g_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_n}{\partial x_1} & \frac{\partial g_n}{\partial x_2} & \dots & \frac{\partial g_n}{\partial x_n} \end{vmatrix} \neq 0$$

for any $x = (x_1, x_2, \dots, x_n)^T$.

Theorem. If Assumptions 1. and 2. are satisfied, then

$$f_Y(y) = \begin{cases} f_X(x) |J(x)|^{-1} & \text{if } y = g(x) \text{ for some } x \\ 0 & \text{otherwise,} \end{cases}$$

where $x = h(y)$.